

Programming Fundamentals 1 – In-Class Test (Solutions with Marks)

Time: 1 Hour

Date: Monday, November 4th, 2025

Section A – Multiple Choice (40 Marks)

Each question = 2 marks.

Q	Question Summary	Correct Answer
A.1	<code>balance = balance + amount;</code>	A – Adds amount to current value
A.2	Constructor rule	A – Name must match class name
A.3	Best variable name	A – <code>ticketPrice</code>
A.4	Constructor parameters	A – Initialize fields
A.5	Boolean literals	A – <code>true</code> and <code>false</code>
A.6	Equality test for ints	B – <code>a == b</code>
A.7	<code>(mark >= 0) && (mark <=100)</code>	A – Between 0 and 100 inclusive
A.8	<code>!(mark < 0)</code>	
A.9	<code>boolean isFriday = false;</code> <code>!isFriday</code>	A – <code>true</code>
A.10	<code>if (canVote) { ... }</code>	A – Prints "You can vote!" when <code>age >= 18</code>
A.11	<code>3 + 6 * 5 - 2</code>	A – <code>31</code> (since 6×5 first $\rightarrow 30 + 3 - 2$)
A.12	<code>if (x = 5)</code>	A – Uses <code>=</code> instead of <code>==</code>
A.13	Teenager or senior	A – <code>!(age >= 13 && age <= 19)</code>
A.14	<code>y *= 3;</code>	B – Multiplies y by 3
A.15	<code>int a=7,b=2; a/b</code>	A – 3
A.16	<code>int a=17,b=5; a % b</code>	A – 2
A.17	<code>int x=4; x-=6; x+=3;</code>	C – $-2 + 3 \rightarrow 1$ (final value)
A.18	Declare array of 10 ints	B – <code>int numbers[] = new int[10];</code>
A.19	<code>numbers.length</code>	B – Total capacity of array
A.20	Array of Strings	D – Both B and C valid

Section B – Long Questions (60 Marks)

Question B.1 – Classes and Objects (15 Marks total)

B.1.1 Constructor (6 Marks)

```
public class TicketMachine {  
    private int price;  
    private int balance;  
    private int total;  
  
    public TicketMachine(int ticketPrice) {  
        price = ticketPrice;  
        balance = 0;  
        total = 0;  
    }  
}
```

B.1.2 Getter for total (4 Marks)

```
public int getTotal() {  
    return total;  
}
```

B.1.3 Setter for price (5 Marks)

```
public void setPrice(int newPrice) {  
    price = newPrice;  
}
```

Question B.2 – Arrays (15 Marks total)

B.2.1 Declare array (2 Marks)

```
double[] wages = new double[20];
```

B.2.2 Use Scanner for input (3 Marks)

```
for (int i = 0; i < wages.length; i++) {  
    wages[i] = input.nextDouble();  
}
```

B.2.3 Calculate and print average (3 Marks)

```
double sum = 0;
for (double w : wages) sum += w;
double average = sum / wages.length;
System.out.println("Average wage = " + average);
```

B.2.4 Print wages over 1000 (3 Marks)

```
for (double w : wages)
    if (w > 1000) System.out.println(w);
```

B.2.5 15% reduction > 1000 (4 Marks)

```
for (int i = 0; i < wages.length; i++) {
    if (wages[i] > 1000)
        wages[i] *= 0.85;
    System.out.println(wages[i]);
}
```

Question B.3 – Selection (15 Marks total)**B.3.1 Print grade method (10 Marks)**

```
public void printGrade(int mark) {
    if (mark >= 70)
        System.out.println("Distinction");
    else if (mark >= 60)
        System.out.println("Merit");
    else if (mark >= 40)
        System.out.println("Pass");
    else
        System.out.println("Fail");
}
```

B.3.2 Validate mark range (5 Marks)

```
public void validateMark(int mark) {
    if (mark >= 0 && mark <= 100)
        System.out.println("The mark is valid");
    else
        System.out.println("The mark is invalid");
}
```

Question B.4 – Loops (15 Marks total)**B.4.1 20 → 10 (3 Marks)**

```
for (int i = 20; i >= 10; i--)  
    System.out.println(i);
```

B.4.2 2 → 17 steps of 3 (3 Marks)

```
for (int i = 2; i <= 17; i += 3)  
    System.out.println(i);
```

B.4.3 Trace loop output (4 Marks)

```
int i = 20;  
while (i <= 27) {  
    System.out.println("Value of i: " + i);  
    i = i + 3;  
}  
System.out.println("Last value of i: " + i);
```

Output

```
Value of i: 20  
Value of i: 23  
Value of i: 26  
Last value of i: 29
```

B.4.4 Nested loops (5 Marks)

```
for (int i = 1; i < 3; i++)  
    for (int j = 0; j < 2; j++)  
        System.out.println("i = " + i + ", j = " + j);
```

Output

```
i = 1, j = 0  
i = 1, j = 1  
i = 2, j = 0  
i = 2, j = 1
```

End of Paper